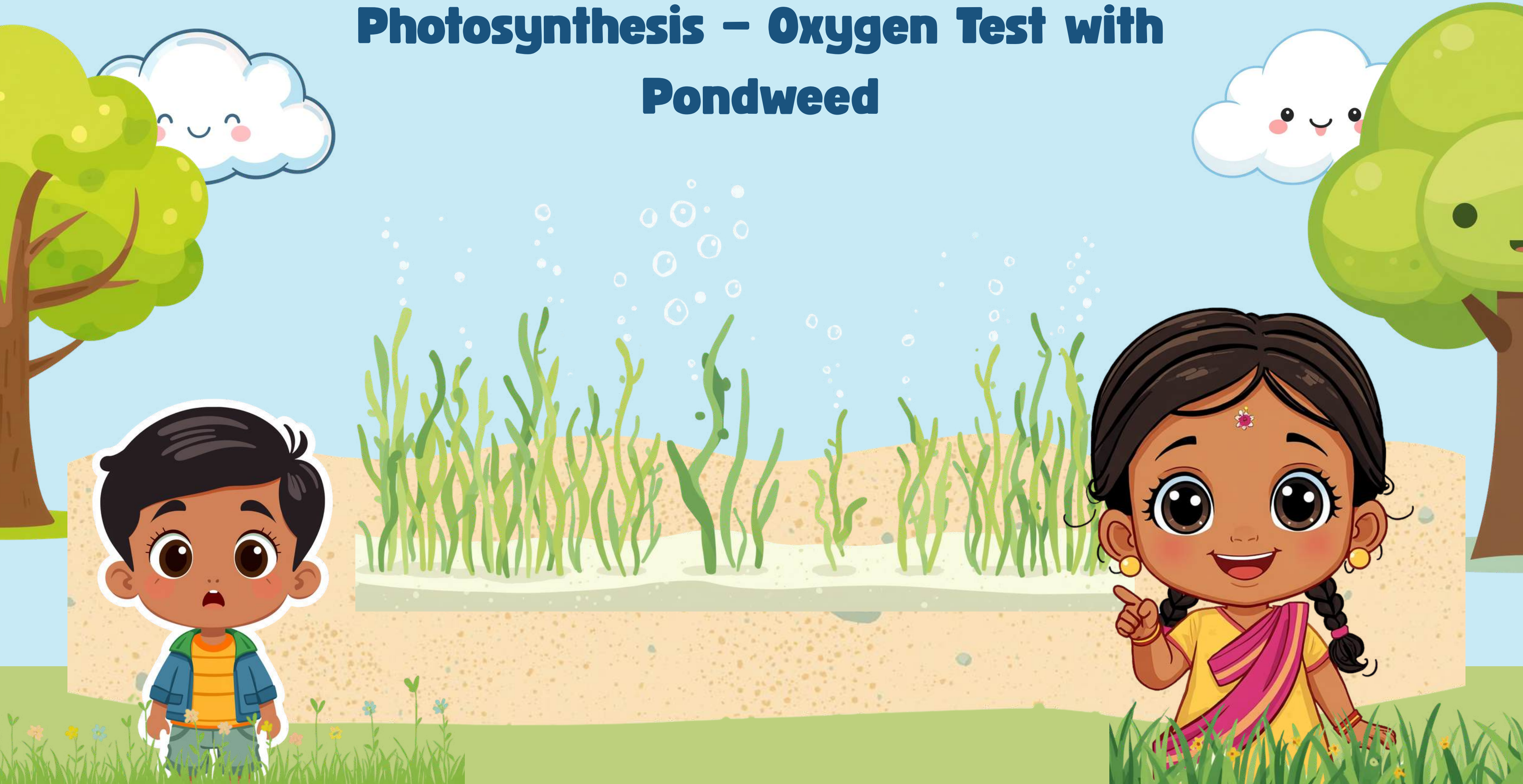


Photosynthesis – Oxygen Test with Pondweed



Learning Objectives

- By the end of today, you will:
- Know how plants make their own food
- Discover what gas plants give us
- See oxygen bubbles from a real plant!



Mystery Bubbles – What Are These Tiny Bubbles?

Can a plant really make air?



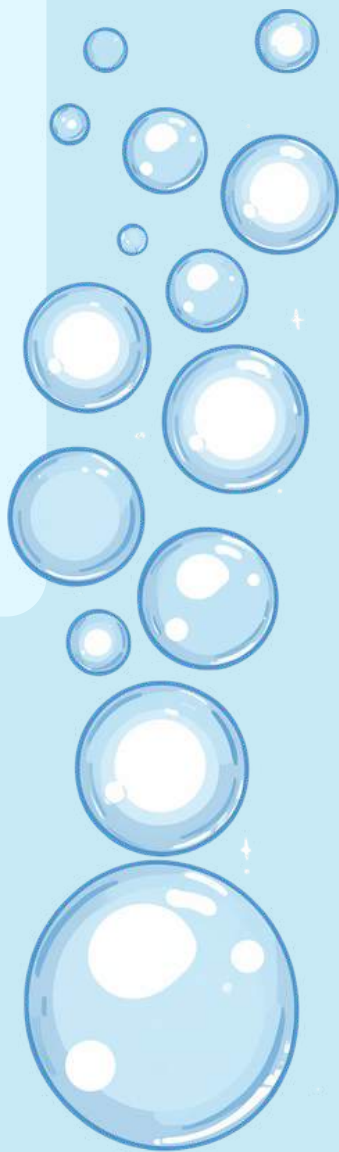


Meet Riya's Discovery

One sunny day, Riya was walking near a pond in Tamil Nadu. She saw tiny bubbles rising from the green plants underwater.

"Is the plant breathing?" Riya asked her teacher.

Her teacher smiled and said, "It is cooking! And making fresh air for all of us!"



What is Photosynthesis?

Photosynthesis = Plant Cooking!

It is the special way a green plant makes its own food. Just like we cook food in our kitchen, plants cook their food using sunlight, water, and air!





The Plant Kitchen Analogy

1

SUNLIGHT = Gas stove (energy to cook!)

2

WATER = From roots (main ingredient)

3

CARBON DIOXIDE + GREEN LEAF = Secret ingredient and the kitchen!

What Does the Plant Make?

Plants make two amazing things:

🍬 **FOOD (Glucose)** - Energy for the plant to grow big and strong!

💨 **OXYGEN** - Fresh air for us to breathe!



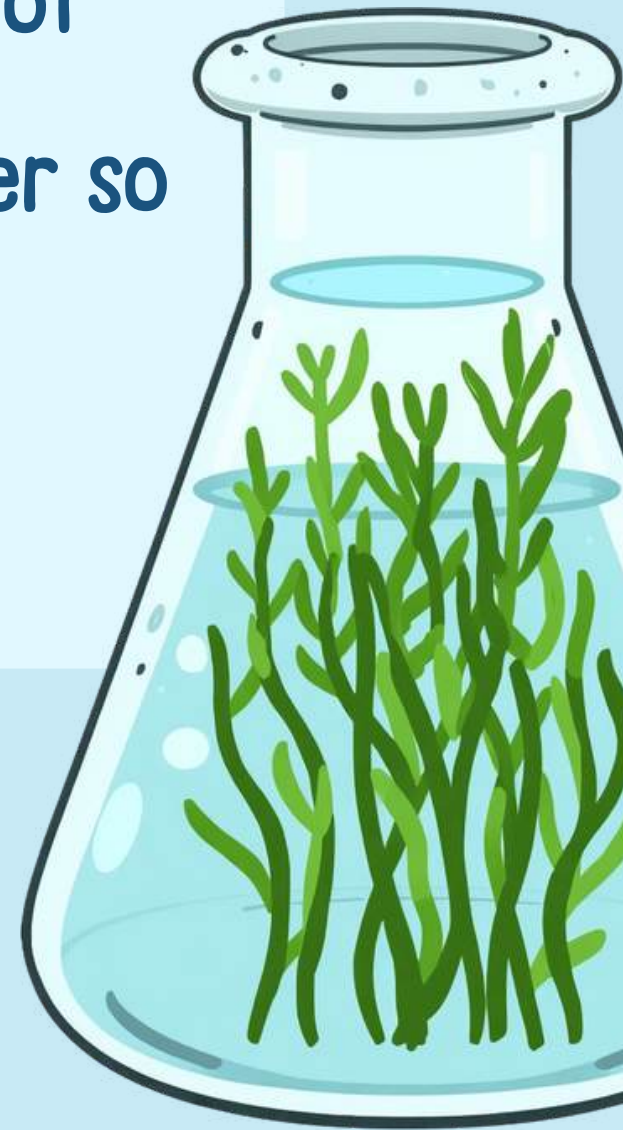
Meet the Pondweed

- Let's meet Hydrilla, our experiment star!
- Hydrilla is a water plant found in Indian ponds
- Special power: We can SEE its photosynthesis!
- It makes oxygen bubbles underwater



The Experiment – Step 1

Place Hydrilla (pondweed) in a clear glass of water. Make sure the plant is fully underwater so we can see the magic happen!



The Experiment – Step 2

Put the glass in bright sunlight near a window or under a lamp. The plant needs sunlight energy to start cooking its food!



The Experiment – Step 3

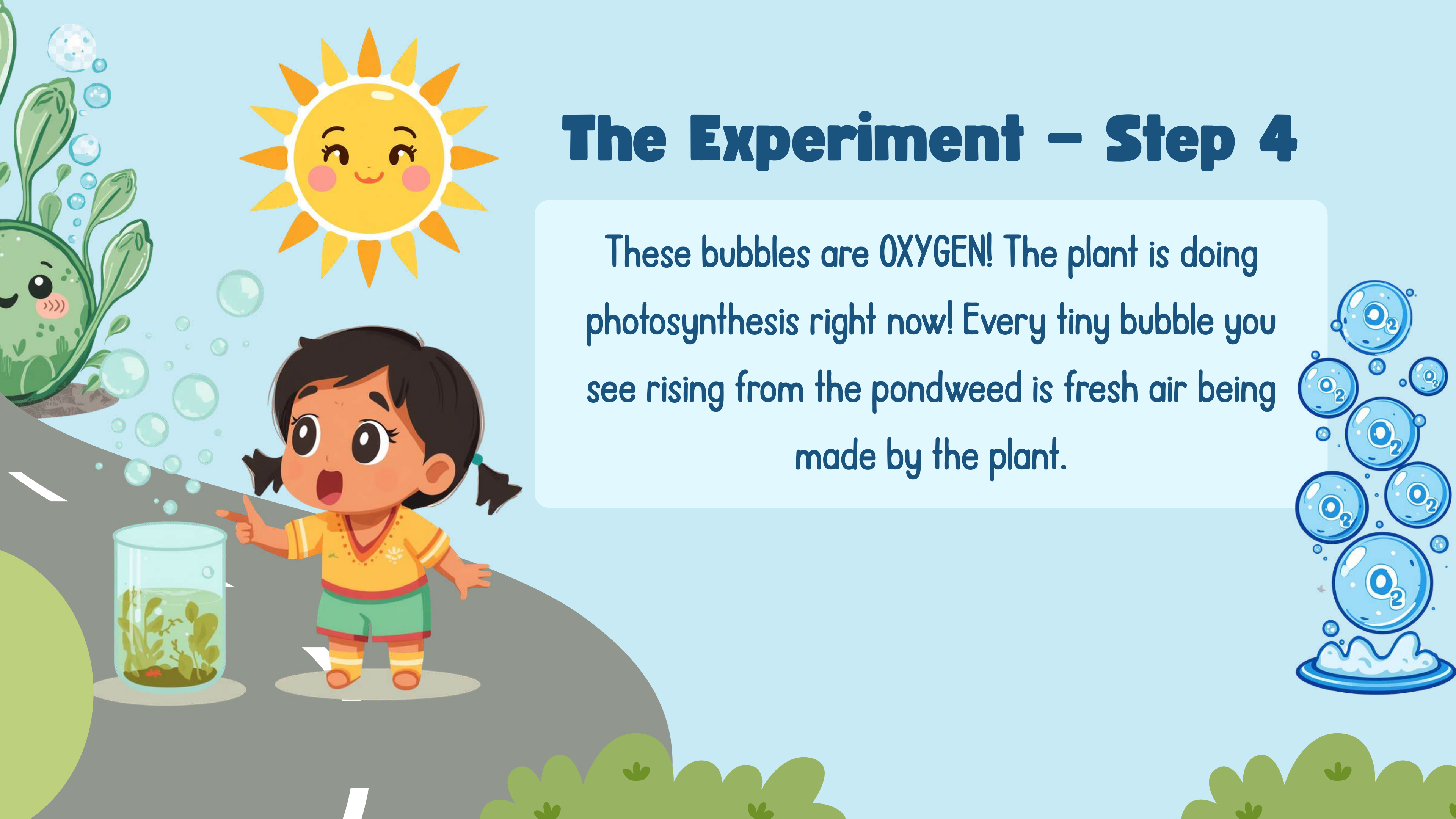
Watch carefully! Tiny bubbles start rising from the plant.

Look closely at the leaves – do you see little round bubbles floating up? Count how many you can spot!



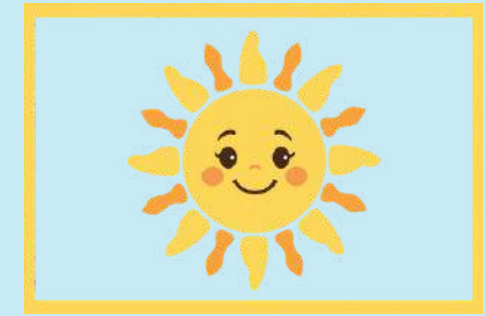
The Experiment – Step 4

These bubbles are OXYGEN! The plant is doing photosynthesis right now! Every tiny bubble you see rising from the pondweed is fresh air being made by the plant.





Sunny vs Dark Comparison



SUNNY JAR
Lots of bubbles!
More photosynthesis!



DARK JAR
No bubbles!
No photosynthesis!



Light is Essential!



Indian Plants Doing Photosynthesis

Plants all around India are making oxygen for us every day!
Here are some plants you might see doing photosynthesis
near your home, school, or garden.

Plants Around Us:



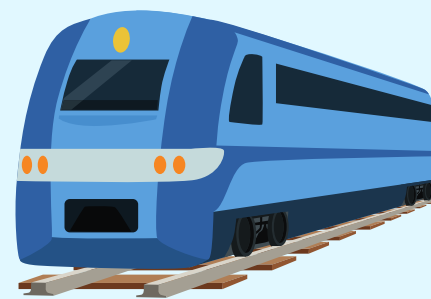
Tulsi



Neem Tree



Banana Leaf



Paddy Fields

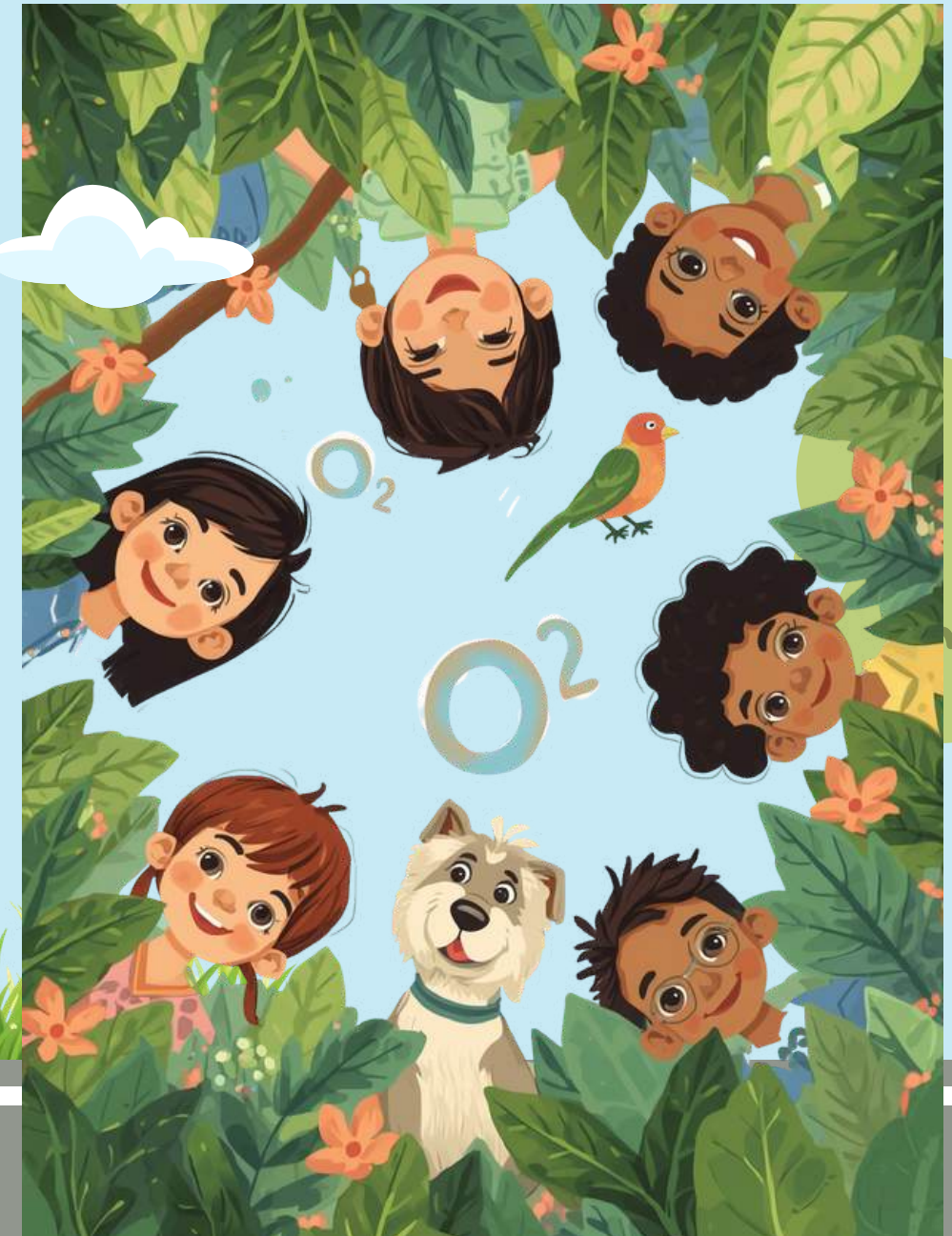


Coconut Tree



Why Oxygen Matters

- Plants are our life-givers!
- Every breath you take is possible because of PLANTS!
- Humans and animals need oxygen to live.
- Plants are our oxygen makers!



Vocabulary Wall

Learn these important words about how plants make food
and help us breathe!

New Words:



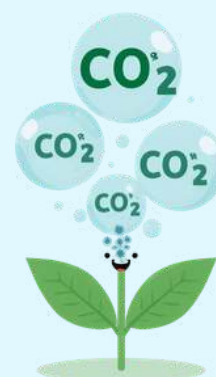
Photosynthesis



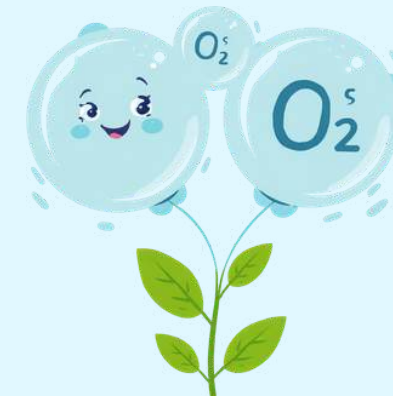
Sunlight



Chlorophyll



Carbon Dioxide



Oxygen





Recap – 3-Word Summary

SUNLIGHT + WATER + AIR $\rightarrow$ OXYGEN + FOOD

Every time you breathe, say THANK YOU to a
plant!

Exit Ticket – Let's Check!

What do we call the way a plant makes its own food?

- a. Photosynthesis
- b. Respiration
- c. Digestion



You learned how plants make food and oxygen through photosynthesis! BONUS Challenge: Name one plant from your home or school doing photosynthesis right now!

Great Job Today!

